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END SEMESTER EXAMINATION - 2025

Name of the Course: B.Tech

Semester: VI

Name of the Paper: Finite Element Method

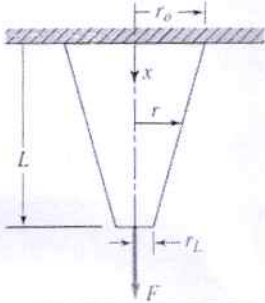
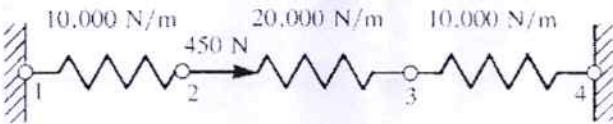
Paper Code: **TME 614**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	What do you understand by discretization finite element method? Discretize this geometry into ten elements. Should all the elements be equal, why? 	CO1, CO2
(b)	What are the general steps for carrying out the finite element procedure.	
(c)	Why does the bar element has four degrees of freedom in 2D plane when it has only two nodes?	
Q2	(20 marks)	
(a)	Derive the element matrix equation for a bar element. What is the relation between its stress and strain?	
(b)	How do you treat non-homogeneous boundary condition?	
(c)	 Determine nodal displacements and reaction force at node 1 and 4.	CO2, CO3
Q3	(20 marks)	
(a)	Derive the T matrix that is used to derive the element stiffness matrix of a bar element in 2D plane.	CO2, CO4
(b)	Determine the displacement at node 1, the reaction force components at nodes 2 and 3, and the element displacements, stresses, and forces. The elements have modulus of elasticity $E = 210 \text{ GPa}$ and cross-sectional areas $A_1 = A_2 = 6 \times 10^4 \text{ m}^2$.	

(c)	Determine the global stiffness matrix and displacements at node 2 and 3.	
Q4	(20 marks)	
(a)	Why beam element has four degrees of freedom and bar element only two?	CO4, CO5
(b)	Explain the characteristics for beam element with diagram.	
(c)	Combine two beam elements in a simply supported beam with a load applied in middle. Derive global stiffness matrix.	
Q5	(20 marks)	
(a)	With diagram derive the finite element formulation of a spring mass system.	CO5, CO6
(b)	Derive the finite element formulation of a spring mass system with two unequal masses connected in series with springs of same stiffness.	
(c)	For spring mass system consisting of two equal masses of 5 kg and connected in series by two springs of stiffness 20N/m. Calculate the two natural frequencies.	

Note for the question paper setters:

- Question paper should cover all the COs of the course.